PROJECT AND TEAM INFORMATION

Project Title

.

SmartBite: Intelligent Food Recommendation and Route Optimization System

Team Name: *Team : DSCPP-III-2026-T076*	
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PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

SmartBite is a C++ console-based food recommendation and route optimization system. It aims to help users choose suitable food places based on factors such as distance, ratings, food preferences, and preparation time. The system brings restaurant discovery and route planning together and includes registration, data handling, and food-service options such as dine-in, takeaway, and delivery. Graphs and Dijkstra's algorithm support route planning, while suitable data structures manage user, restaurant, menu, and order information. Travel time and payment are simulated; the project does not use real-time GPS or live payment services.

Updated Project Approach and Architecture

The project is being developed in C++ using data structures and algorithms in VS Code. The current work is organized into Registration, Data, Graph, and Dining modules. Registration handles the user access flow; Data organizes application information; Graph supports location representation and route calculation; and Dining connects the food-service flow. The intended design uses graph representations and Dijkstra's algorithm for shortest-path calculation, with vectors, queues, stacks, linked lists, and hash maps where appropriate for storing and processing information. Search and sorting methods support restaurant filtering and ranking. The application is a local console prototype: travel estimates, restaurant details, crowd information, and payment behavior are simulated rather than retrieved from live external services.

Tasks Completed

Task Completed	Team Member
Registration module development and integration	Komal Singh
Data module work and data organization	Dhairya Bhardwaj
Graph module and route-planning logic	Nimish Sharma
Dining module flow development	Dhairya Bhardwaj and team

Challenges/Roadblocks

The main challenge is coordinating separate modules so that data passed between them uses consistent structures and interfaces. Route calculation also depends on correctly representing locations, edges, and weights in the graph. Integration can expose compiler warnings or mismatches between module assumptions. The team is addressing these issues by testing modules individually, checking function interfaces and data types, and then validating the complete user flow. Another limitation is that restaurant, crowd, travel-time, and payment information is simulated; therefore, results should be presented as a prototype demonstration rather than a real-time service.

Tasks Pending

Task Pending	Team Member (to complete the task)
Complete cross-module integration and regression testing	All team members

Project Outcome/Deliverables

The intended deliverable is an integrated C++ console application that supports user registration, food and restaurant data handling, route planning, and dine-in, takeaway, and delivery flows. The project applies data structures and algorithms to restaurant selection and shortest-route calculation. Final deliverables also include source code, testing evidence, and project documentation. The prototype simulates travel details and payments and does not claim live GPS or online payment integration.

Progress Overview

Phase-II development has progressed across four modules: Registration, Data, Graph, and Dining. The Dining flow has been developed through its main interaction path, and the modules are being prepared for integrated validation. The exact overall completion percentage has not been measured, so no percentage is claimed here. Remaining work includes checking the complete integrated flow, resolving any remaining compiler or interface issues, recording test outcomes, capturing output screenshots, and verifying repository and commit details before submission.

Codebase Information

Project: SmartBite — Intelligent Food Recommendation and Route Optimization System. Team ID: DSCPP-III-2026-T076. The source is maintained as a C++ project. Repository URL, active branch, and key commit hashes should be inserted from the team's actual GitHub repository before submission; these details were not available in the provided materials and have not been invented.

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
regestration and password module	pass	
graph module	pass / progress	
dining module	pass / improvement	
data module	pass	

Deliverables Progress

1. Module Development:** Registration, Data, Graph, and Dining are the key Phase-II modules currently under development.
2. Dining Module:** Work has been carried out on the Dining user flow. However, the complete integration status of all modules still needs confirmation.
3. Route Planning Validation:** The route-planning functionality should be tested using known graph inputs and expected shortest paths to verify the correctness of the results.
4. Registration and Data Validation:** Registration and data operations should be checked using both valid and invalid inputs to ensure proper handling.
5. End-to-End Testing:** The dine-in, takeaway, and delivery workflows should be tested from start to finish.
6. Test Results and Screenshots:** Actual test results, pass/fail status, and screenshots should be added after executing the tests. Since these results are not available in the source materials, no testing success claims are made. .